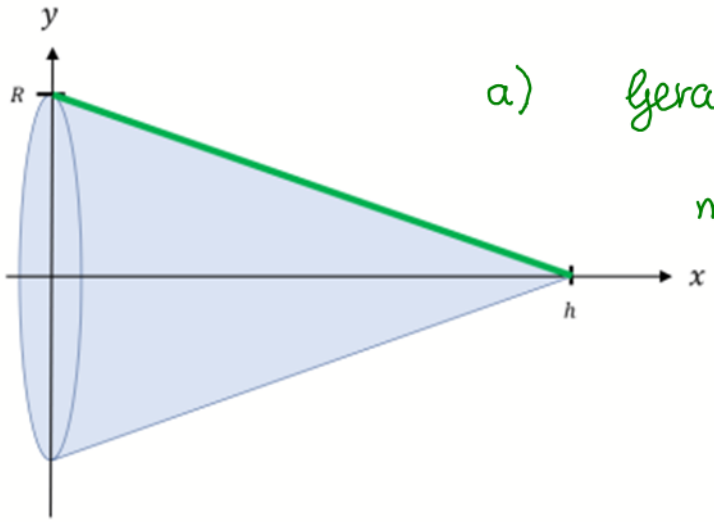
Lösung: Kegelvolumen als Rotationsvolumen



a) Geradengleichung: $f(x) = mx + b$

mit $f(0) = R \iff b = R$

$f(h) = 0 \iff m \cdot h + R = 0$

$\iff m = -R/h$

$\Rightarrow \boxed{f(x) = -\frac{R}{h} \cdot x + R}$

b) $V = \pi \cdot \int_0^h (f(x))^2 \, dx$

$= \pi \cdot \int_0^h (R - \frac{R}{h} x)^2 \, dx = \pi \int_0^h (R^2 - 2\frac{R^2}{h} x + \frac{R^2}{h^2} x^2) \, dx$

$= \pi \cdot \left[R^2 x - \frac{R^2 x^2}{h} + \frac{R^2 x^3}{3h^2} \right]_0^h = \pi \cdot \left(R^2 h - \frac{R^2 h^2}{h} + \frac{R^2 h^3}{3h^2} \right)$

$= \pi \cdot \left(R^2 h - R^2 h + \frac{1}{3} R^2 h \right) = \frac{1}{3} \pi R^2 h$

c) $V = \frac{1}{3} G \cdot h = \frac{1}{3} \pi R^2 h$ ✓